

# gocachemark - Go Cache Benchmark Report @ 2026-01-02 12:10:56 EST

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[gocachemark](#) · darwin/arm64 · 16 CPUs · go1.25.5

2026-01-02 12:10:56 EST gocachemark

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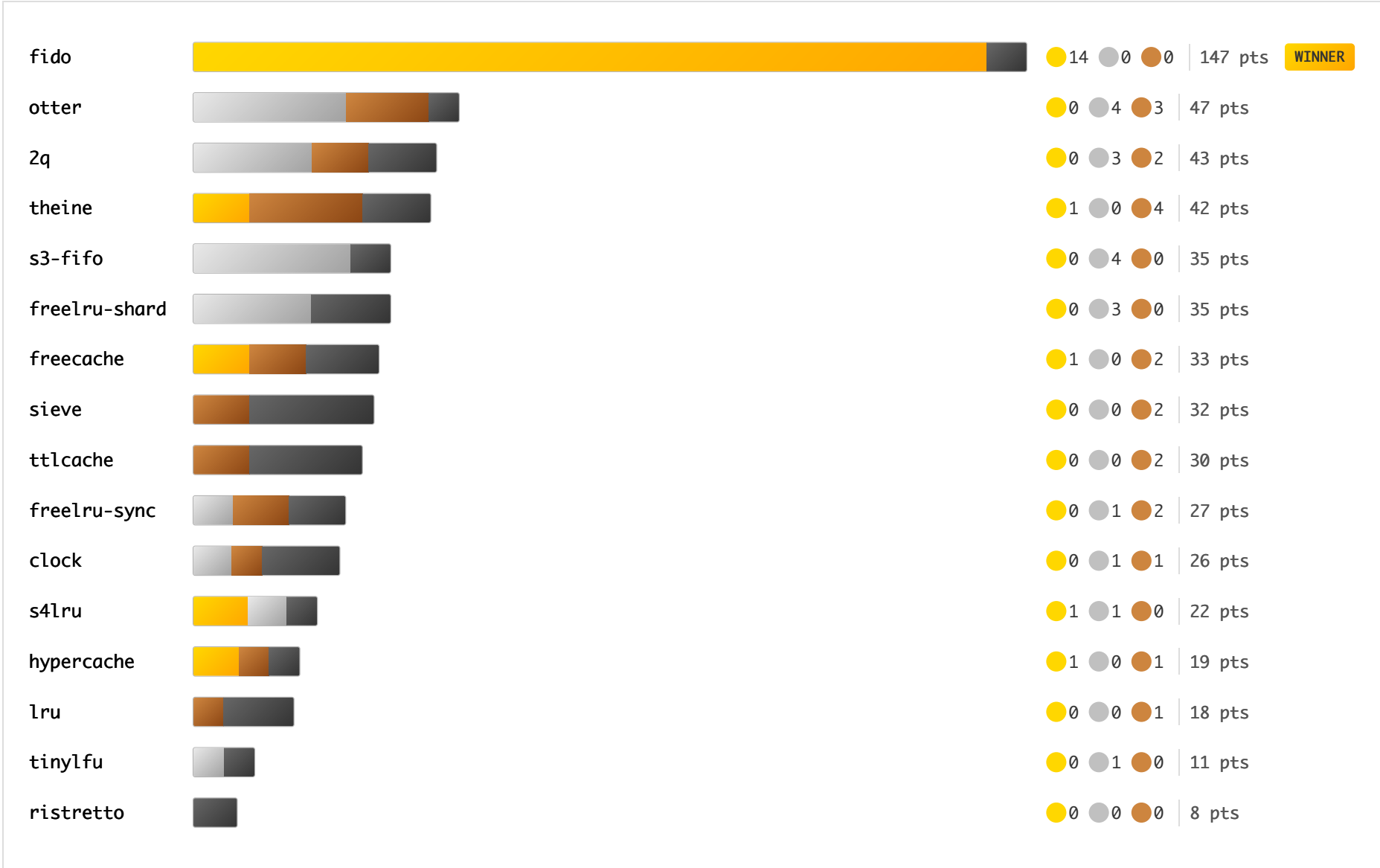
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## Medal Standings

Ranked by total points earned across all benchmarks

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Medals by Category

Hit Rate — Winner: **fido**

|              |         |
|--------------|---------|
| fido         | ● 6     |
| s4lru        | ● 1 ● 1 |
| theine       | ● 1 ● 2 |
| freecache    | ● 1     |
| s3-fifo      | ● 4     |
| 2q           | ● 3 ● 2 |
| tinylfu      | ● 1     |
| sieve        | ● 2     |
| lru          | ● 1     |
| ttlcache     | ● 1     |
| freelru-sync | ● 1     |
| otter        | ● 1     |
| clock        | ● 1     |

| Benchmark     | Gold      | Silver  | Bronze                      |
|---------------|-----------|---------|-----------------------------|
| CDN           | fido      | s3-fifo | 2q                          |
| Meta          | fido      | s3-fifo | theine                      |
| Zipf          | theine    | tinylfu | otter                       |
| Twitter       | fido      | 2q      | sieve                       |
| Wikipedia     | fido      | s3-fifo | theine                      |
| Thesios Block | s4lru     | 2q      | sieve                       |
| Thesios File  | fido      | s4lru   | lru, ttlcache, freelru-sync |
| IBM Docker    | freecache | 2q      | clock                       |
| Tencent Photo | fido      | s3-fifo | 2q                          |

Latency — Winner: **fido**

|               |         |
|---------------|---------|
| fido          | ● 3     |
| otter         | ● 1 ● 1 |
| clock         | ● 1     |
| freelru-shard | ● 1     |
| freelru-sync  | ● 1     |
| freecache     | ● 1     |

| Benchmark   | Gold | Silver        | Bronze       |
|-------------|------|---------------|--------------|
| String Keys | fido | clock         | freelru-sync |
| Int Keys    | fido | freelru-shard | otter        |
| GetOrSet    | fido | otter         | freecache    |

Throughput — Winner: **fido**

|               |     |
|---------------|-----|
| fido          | ● 5 |
| otter         | ● 3 |
| freelru-shard | ● 2 |
| theine        | ● 2 |
| freecache     | ● 1 |
| ttlcache      | ● 1 |
| hypercache    | ● 1 |

| Benchmark  | Gold | Silver        | Bronze     |
|------------|------|---------------|------------|
| String Get | fido | otter         | theine     |
| String Set | fido | freelru-shard | freecache  |
| Int Get    | fido | otter         | theine     |
| Int Set    | fido | freelru-shard | ttlcache   |
| GetOrSet   | fido | otter         | hypercache |

### Memory — Winner: **hypercache**

|              |     |
|--------------|-----|
| hypercache   | ● 1 |
| freelru-sync | ● 1 |
| otter        | ● 1 |

| Benchmark | Gold       | Silver       | Bronze |
|-----------|------------|--------------|--------|
| Overhead  | hypercache | freelru-sync | otter  |

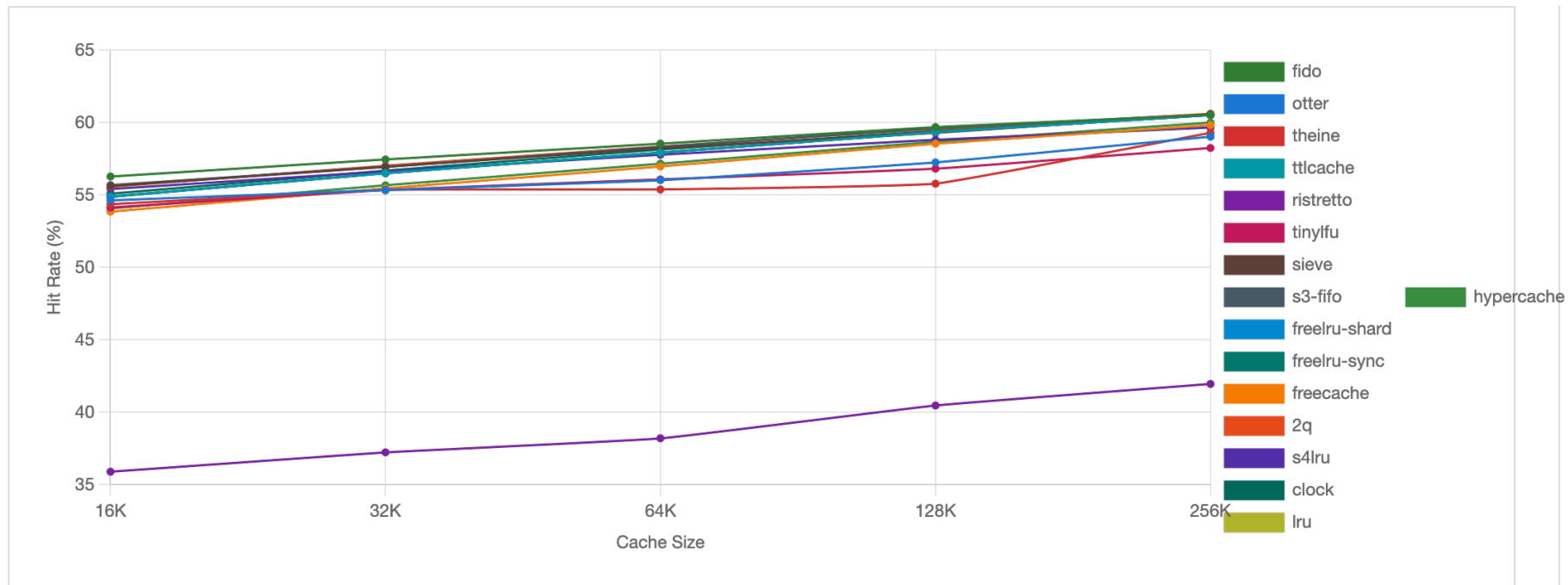
## Hit Rate Benchmarks



Higher hit rates mean better cache effectiveness. Tests measure how often requested items are found in the cache across different cache sizes.

### CDN Production Trace

2M operations, ~768K unique keys. Source: [libCacheSim](#)



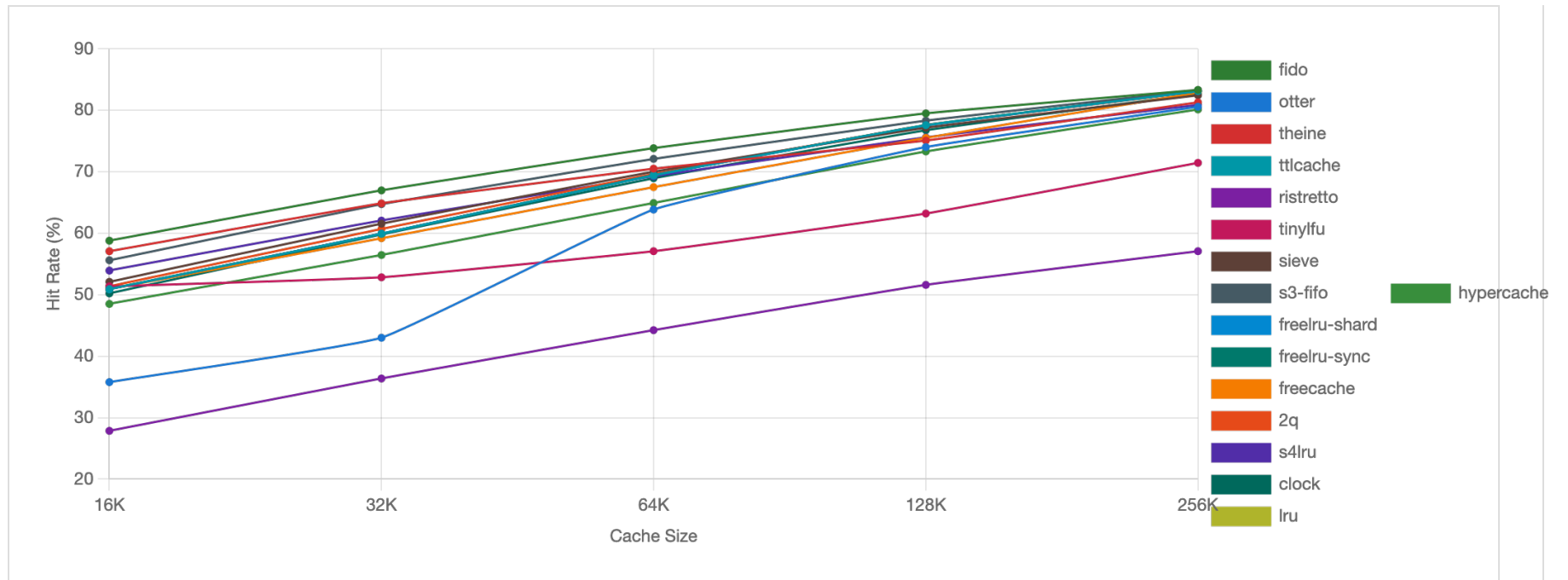
Results Table

| Cache        | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|--------------|---------|---------|---------|---------|---------|---------|
| fido         | 56.282% | 57.457% | 58.553% | 59.694% | 60.554% | 58.508% |
| s3-fifo      | 55.690% | 56.948% | 58.327% | 59.611% | 60.554% | 58.226% |
| 2q           | 55.575% | 57.019% | 58.349% | 59.559% | 60.613% | 58.223% |
| sieve        | 55.634% | 56.947% | 58.248% | 59.298% | 60.587% | 58.143% |
| clock        | 55.093% | 56.675% | 58.138% | 59.422% | 60.566% | 57.979% |
| freelru-sync | 54.901% | 56.516% | 57.944% | 59.345% | 60.533% | 57.848% |
| lru          | 54.901% | 56.516% | 57.944% | 59.345% | 60.533% | 57.848% |
| ttlcache     | 54.901% | 56.516% | 57.944% | 59.345% | 60.533% | 57.848% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| freelru-shard | 54.886% | 56.507% | 57.940% | 59.344% | 60.532% | 57.842% |
| s4lru         | 55.423% | 56.644% | 57.789% | 58.820% | 59.674% | 57.670% |
| hypercache    | 54.083% | 55.685% | 57.168% | 58.675% | 60.011% | 57.124% |
| freecache     | 53.858% | 55.457% | 56.976% | 58.548% | 59.833% | 56.934% |
| otter         | 54.638% | 55.351% | 56.026% | 57.264% | 59.038% | 56.463% |
| tinylfu       | 54.139% | 55.350% | 56.081% | 56.825% | 58.248% | 56.128% |
| theine        | 54.365% | 55.356% | 55.385% | 55.795% | 59.299% | 56.040% |
| ristretto     | 35.908% | 37.239% | 38.214% | 40.482% | 41.973% | 38.763% |

### Meta KVCache Production Trace

5M operations. Source: [CacheLib](#)



Results Table

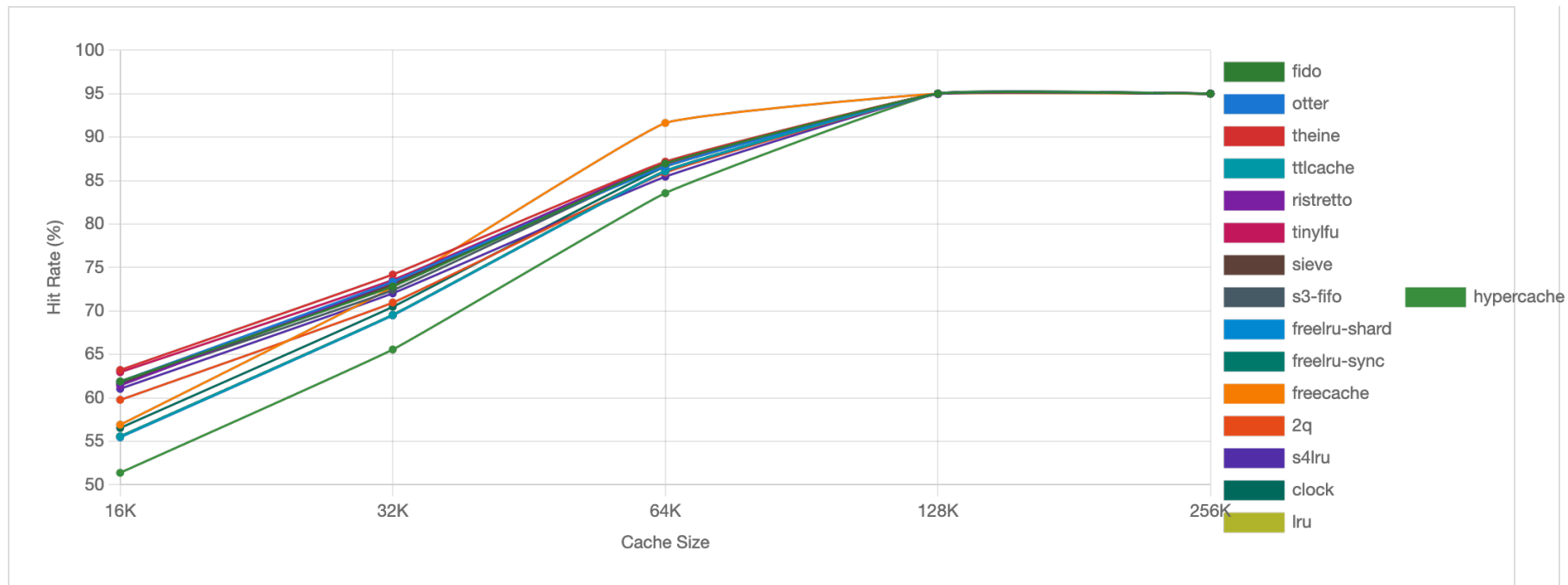
| Cache        | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|--------------|---------|---------|---------|---------|---------|---------|
| fido         | 58.798% | 66.991% | 73.854% | 79.505% | 83.330% | 72.495% |
| s3-fifo      | 55.609% | 64.742% | 72.101% | 78.334% | 83.275% | 70.812% |
| theine       | 57.070% | 64.868% | 70.503% | 75.080% | 81.288% | 69.762% |
| sieve        | 52.098% | 61.564% | 70.016% | 77.166% | 82.451% | 68.659% |
| 2q           | 51.375% | 60.714% | 69.647% | 77.533% | 83.073% | 68.469% |
| s4lru        | 53.943% | 62.068% | 69.274% | 75.591% | 80.867% | 68.349% |
| freelru-sync | 50.934% | 59.924% | 69.436% | 77.579% | 83.116% | 68.198% |
| lru          | 50.934% | 59.924% | 69.436% | 77.579% | 83.116% | 68.198% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| ttlcache      | 50.934% | 59.924% | 69.436% | 77.579% | 83.116% | 68.198% |
| freelru-shard | 50.895% | 59.941% | 69.400% | 77.509% | 83.114% | 68.172% |
| clock         | 50.228% | 59.802% | 68.951% | 76.758% | 82.677% | 67.683% |
| freecache     | 50.926% | 59.195% | 67.513% | 75.525% | 82.894% | 67.211% |
| hypercache    | 48.525% | 56.469% | 64.921% | 73.303% | 80.145% | 64.672% |
| otter         | 35.776% | 43.001% | 63.864% | 74.024% | 80.578% | 59.449% |
| tinylfu       | 51.354% | 52.830% | 57.071% | 63.202% | 71.448% | 59.181% |
| ristretto     | 27.852% | 36.374% | 44.235% | 51.599% | 57.067% | 43.425% |

### Zipf Synthetic Trace

alpha=0.8, 2M operations, 100K keyspace





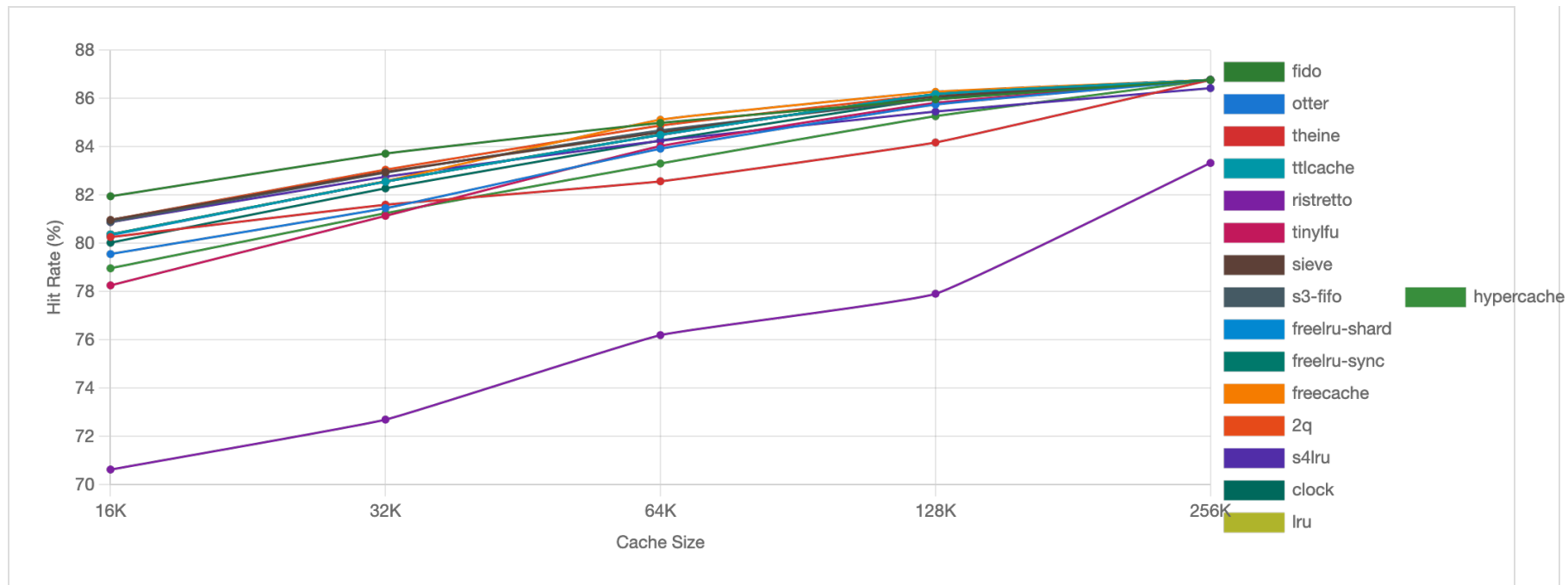
Results Table

| Cache     | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|-----------|---------|---------|---------|---------|---------|---------|
| theine    | 63.200% | 74.187% | 87.175% | 95.014% | 95.014% | 82.918% |
| tinylfu   | 62.964% | 73.573% | 86.917% | 95.014% | 95.014% | 82.696% |
| otter     | 61.887% | 73.369% | 86.626% | 95.014% | 95.014% | 82.382% |
| ristretto | 61.420% | 73.312% | 87.084% | 94.944% | 95.007% | 82.353% |
| fido      | 61.896% | 72.833% | 86.965% | 95.014% | 95.014% | 82.344% |
| sieve     | 61.558% | 73.071% | 86.998% | 95.014% | 95.014% | 82.331% |
| freecache | 56.915% | 72.737% | 91.628% | 95.014% | 95.014% | 82.261% |
| s3-fifo   | 61.740% | 72.401% | 86.912% | 95.014% | 95.014% | 82.216% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| s4lru         | 61.043% | 72.030% | 85.442% | 95.014% | 95.014% | 81.709% |
| 2q            | 59.772% | 70.969% | 85.935% | 95.014% | 95.014% | 81.341% |
| clock         | 56.541% | 70.476% | 86.600% | 95.014% | 95.014% | 80.729% |
| freelru-sync  | 55.569% | 69.547% | 86.112% | 95.014% | 95.014% | 80.251% |
| ttlcache      | 55.569% | 69.547% | 86.112% | 95.014% | 95.014% | 80.251% |
| lru           | 55.569% | 69.547% | 86.112% | 95.014% | 95.014% | 80.251% |
| freelru-shard | 55.472% | 69.499% | 86.069% | 95.014% | 95.014% | 80.214% |
| hypercache    | 51.380% | 65.560% | 83.560% | 95.014% | 95.014% | 78.106% |

## Twitter Production Cache Trace

2M operations, cluster001+cluster052. Source: [Twitter Cache Trace](#)



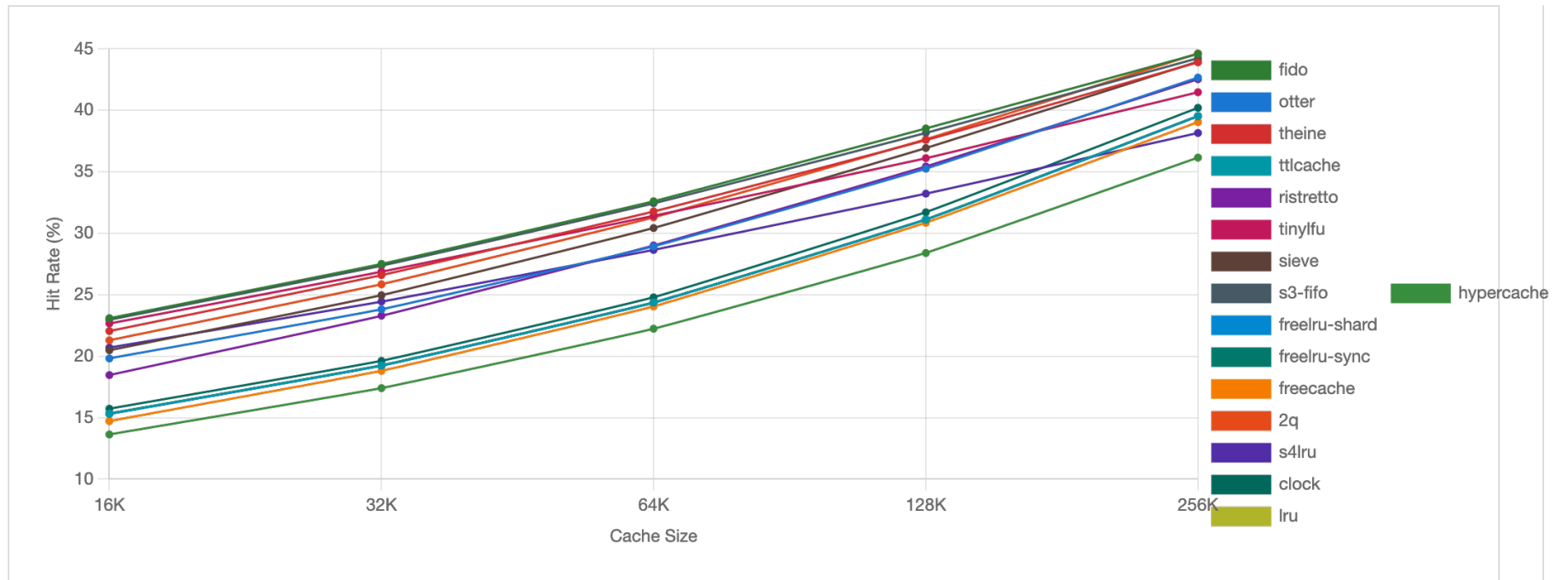
Results Table

| Cache        | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|--------------|---------|---------|---------|---------|---------|---------|
| fido         | 81.946% | 83.717% | 84.991% | 85.969% | 86.773% | 84.679% |
| 2q           | 80.970% | 83.050% | 84.875% | 86.166% | 86.773% | 84.367% |
| sieve        | 80.971% | 82.965% | 84.599% | 86.077% | 86.773% | 84.277% |
| s3-fifo      | 80.891% | 82.933% | 84.684% | 85.969% | 86.773% | 84.250% |
| freecache    | 80.357% | 82.578% | 85.120% | 86.278% | 86.773% | 84.221% |
| freelru-sync | 80.371% | 82.561% | 84.500% | 86.182% | 86.773% | 84.077% |
| lru          | 80.371% | 82.561% | 84.500% | 86.182% | 86.773% | 84.077% |
| ttlcache     | 80.371% | 82.561% | 84.500% | 86.182% | 86.773% | 84.077% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| freelru-shard | 80.341% | 82.561% | 84.492% | 86.179% | 86.772% | 84.069% |
| s4lru         | 80.883% | 82.760% | 84.250% | 85.464% | 86.425% | 83.957% |
| clock         | 80.030% | 82.285% | 84.257% | 85.997% | 86.773% | 83.868% |
| otter         | 79.559% | 81.460% | 83.924% | 85.746% | 86.773% | 83.492% |
| tinylfu       | 78.260% | 81.144% | 84.030% | 85.820% | 86.772% | 83.205% |
| hypercache    | 78.974% | 81.245% | 83.314% | 85.275% | 86.773% | 83.116% |
| theine        | 80.260% | 81.599% | 82.566% | 84.176% | 86.770% | 83.074% |
| ristretto     | 70.628% | 72.697% | 76.199% | 77.914% | 83.327% | 76.153% |

## Wikipedia CDN Upload Trace

2M operations, upload.wikimedia.org. Source: [libCacheSim](#)



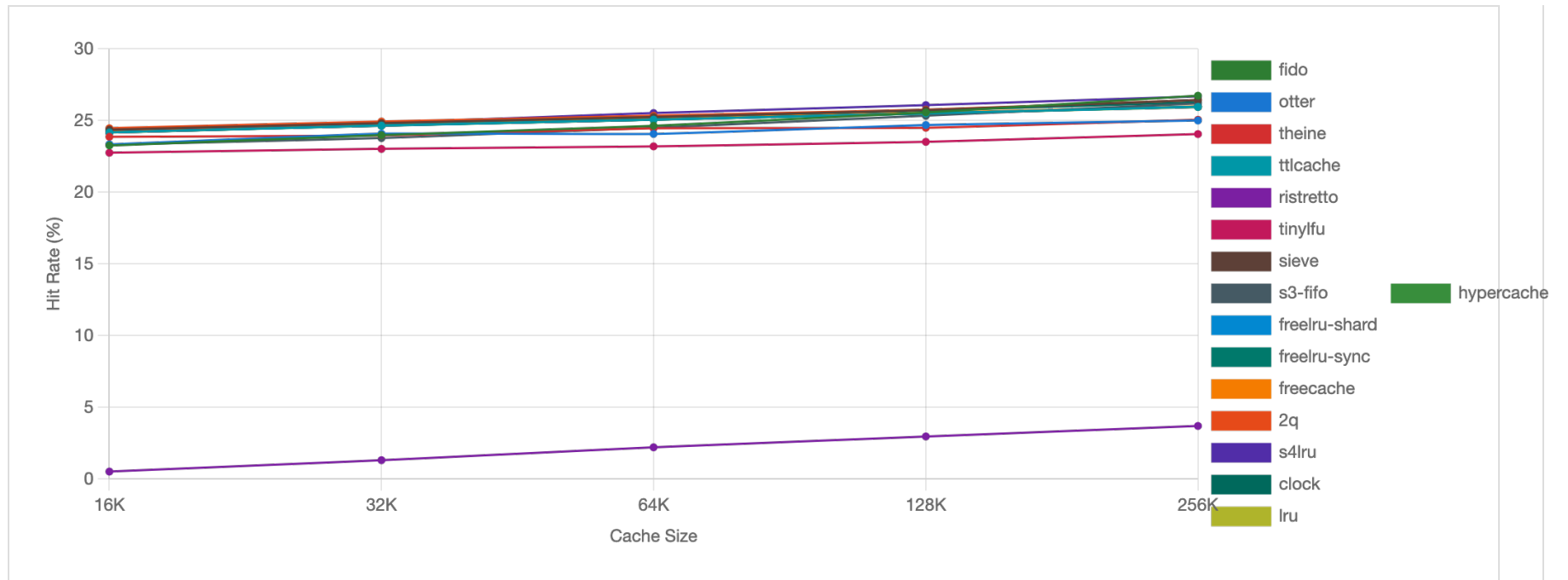
Results Table

| Cache     | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|-----------|---------|---------|---------|---------|---------|---------|
| fido      | 23.093% | 27.506% | 32.610% | 38.534% | 44.601% | 33.269% |
| s3-fifo   | 22.970% | 27.374% | 32.441% | 38.173% | 44.250% | 33.042% |
| theine    | 22.055% | 26.595% | 31.783% | 37.577% | 43.910% | 32.384% |
| 2q        | 21.286% | 25.850% | 31.285% | 37.635% | 44.619% | 32.135% |
| tinylfu   | 22.663% | 26.880% | 31.434% | 36.112% | 41.483% | 31.714% |
| sieve     | 20.476% | 24.956% | 30.430% | 36.932% | 43.963% | 31.351% |
| otter     | 19.825% | 23.807% | 28.928% | 35.247% | 42.655% | 30.092% |
| ristretto | 18.473% | 23.291% | 29.012% | 35.435% | 42.538% | 29.750% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| s4lru         | 20.697% | 24.438% | 28.654% | 33.215% | 38.155% | 29.032% |
| clock         | 15.725% | 19.617% | 24.792% | 31.711% | 40.196% | 26.408% |
| freelru-sync  | 15.327% | 19.249% | 24.375% | 31.121% | 39.530% | 25.920% |
| ttlcache      | 15.327% | 19.249% | 24.375% | 31.121% | 39.530% | 25.920% |
| lru           | 15.327% | 19.249% | 24.375% | 31.121% | 39.530% | 25.920% |
| freelru-shard | 15.299% | 19.233% | 24.364% | 31.117% | 39.529% | 25.908% |
| freecache     | 14.725% | 18.806% | 24.046% | 30.859% | 39.044% | 25.496% |
| hypercache    | 13.633% | 17.398% | 22.230% | 28.402% | 36.146% | 23.562% |

## Google Thesios I/O Block Trace

400K read operations, ~322K unique blocks. Source: [Tectonic-Shift \(SOSP '24\)](#)



Results Table

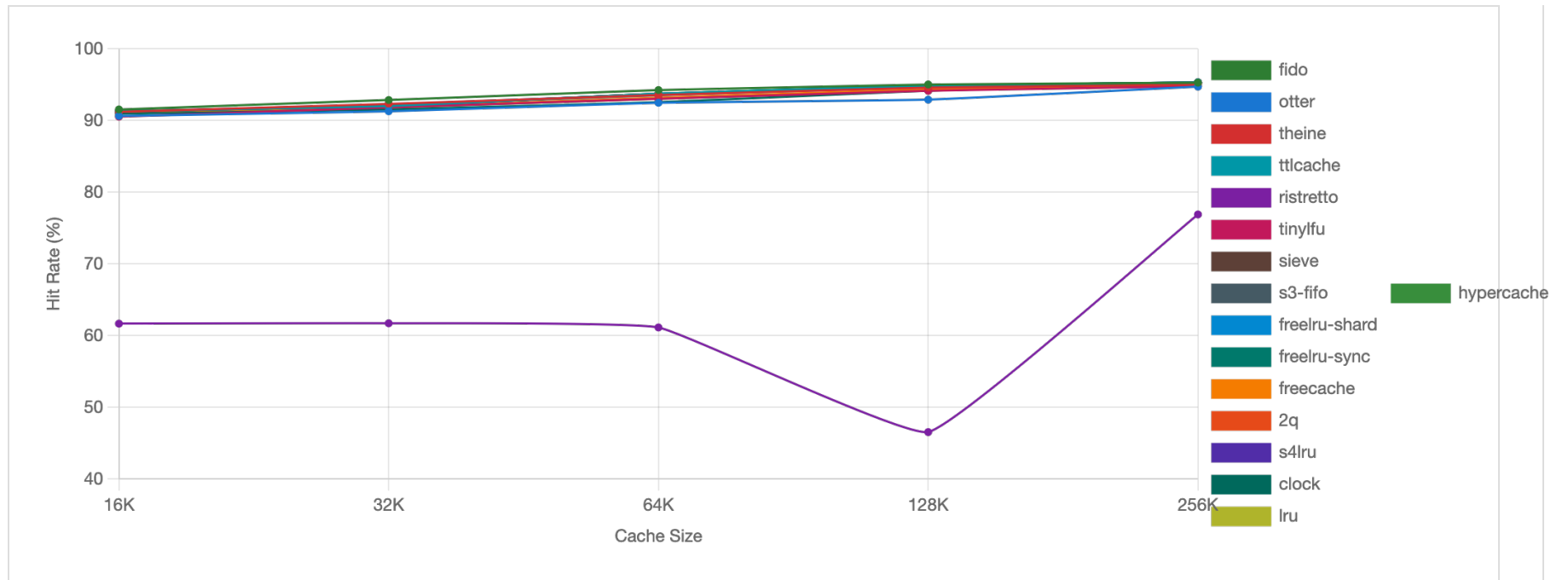
| Cache        | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|--------------|---------|---------|---------|---------|---------|---------|
| s4lru        | 24.191% | 24.839% | 25.526% | 26.082% | 26.693% | 25.466% |
| 2q           | 24.467% | 24.925% | 25.340% | 25.760% | 26.430% | 25.384% |
| sieve        | 24.362% | 24.818% | 25.246% | 25.726% | 26.428% | 25.316% |
| clock        | 24.278% | 24.756% | 25.200% | 25.694% | 26.280% | 25.242% |
| freecache    | 24.210% | 24.685% | 25.103% | 25.537% | 25.991% | 25.105% |
| freelru-sync | 24.184% | 24.656% | 25.067% | 25.501% | 25.966% | 25.075% |
| lru          | 24.184% | 24.656% | 25.067% | 25.501% | 25.966% | 25.075% |
| ttlcache     | 24.184% | 24.656% | 25.067% | 25.501% | 25.966% | 25.075% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| freelru-shard | 24.177% | 24.653% | 25.067% | 25.501% | 25.965% | 25.072% |
| hypercache    | 24.168% | 24.640% | 25.059% | 25.492% | 25.940% | 25.060% |
| fido          | 23.262% | 23.995% | 24.636% | 25.573% | 26.738% | 24.841% |
| s3-fifo       | 23.308% | 23.784% | 24.505% | 25.336% | 26.186% | 24.624% |
| theine        | 23.870% | 23.970% | 24.460% | 24.496% | 25.060% | 24.371% |
| otter         | 23.341% | 24.091% | 24.068% | 24.689% | 25.014% | 24.240% |
| tinylfu       | 22.756% | 23.028% | 23.198% | 23.519% | 24.064% | 23.313% |
| ristretto     | 0.529%  | 1.321%  | 2.217%  | 2.967%  | 3.714%  | 2.150%  |

### Google Thesios I/O File Trace

400K read operations, ~46K unique files. Source: [Tectonic-Shift \(SOSP '24\)](#)





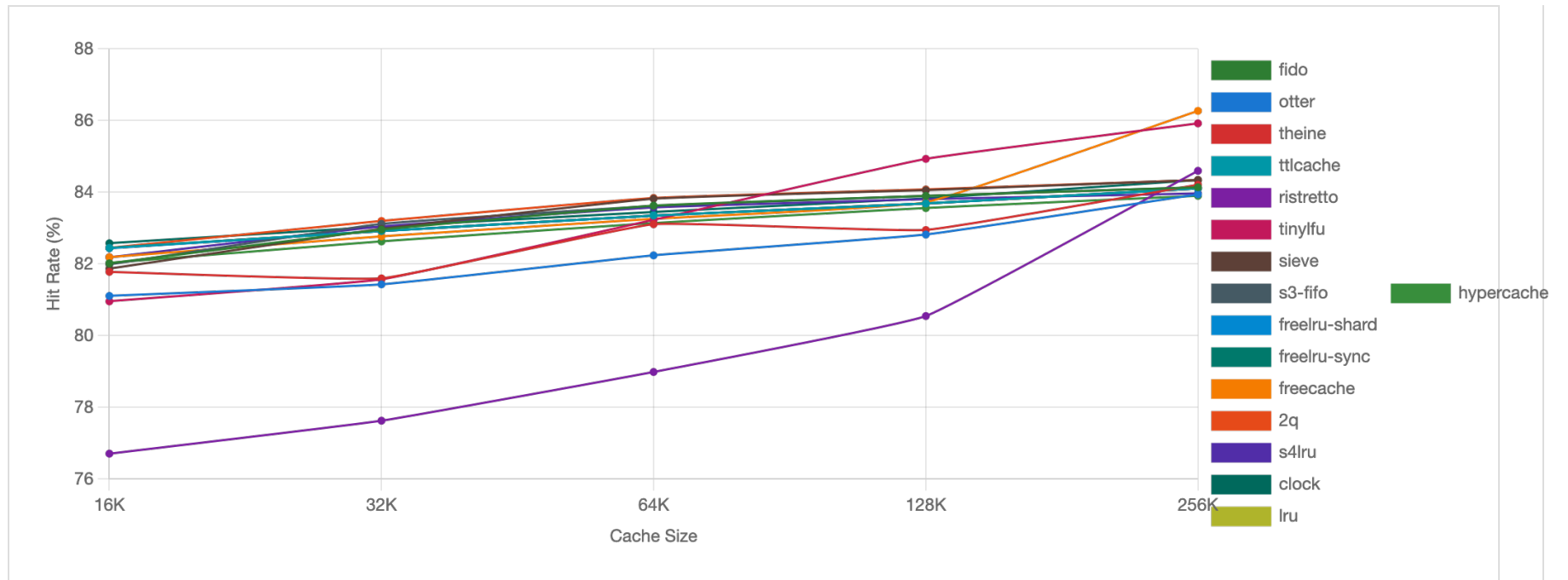
Results Table

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| fido          | 91.553% | 92.881% | 94.249% | 95.030% | 95.302% | 93.803% |
| s4lru         | 91.260% | 92.211% | 93.663% | 94.840% | 95.310% | 93.457% |
| lru           | 91.295% | 92.027% | 93.732% | 94.855% | 95.311% | 93.444% |
| ttlcache      | 91.295% | 92.027% | 93.732% | 94.855% | 95.311% | 93.444% |
| freelru-sync  | 91.295% | 92.027% | 93.732% | 94.855% | 95.311% | 93.444% |
| freelru-shard | 91.291% | 92.026% | 93.730% | 94.849% | 95.310% | 93.441% |
| sieve         | 91.287% | 92.041% | 93.699% | 94.840% | 95.260% | 93.425% |
| 2q            | 91.279% | 92.094% | 93.689% | 94.714% | 95.290% | 93.413% |

| Cache      | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|------------|---------|---------|---------|---------|---------|---------|
| s3-fifo    | 91.193% | 92.181% | 93.520% | 94.671% | 95.211% | 93.355% |
| theine     | 91.230% | 92.334% | 93.568% | 94.603% | 95.025% | 93.352% |
| freecache  | 91.228% | 91.896% | 93.193% | 94.357% | 95.148% | 93.164% |
| hypercache | 91.190% | 91.827% | 93.043% | 94.214% | 94.912% | 93.037% |
| clock      | 90.965% | 91.559% | 92.565% | 94.175% | 95.170% | 92.887% |
| tinylfu    | 90.536% | 91.832% | 93.032% | 94.147% | 94.779% | 92.865% |
| otter      | 90.636% | 91.297% | 92.483% | 92.925% | 94.738% | 92.416% |
| ristretto  | 61.689% | 61.725% | 61.143% | 46.561% | 76.903% | 61.604% |

## IBM Docker Registry Trace

725K GET requests, ~121K unique URIs. Source: [FAST '18](#)



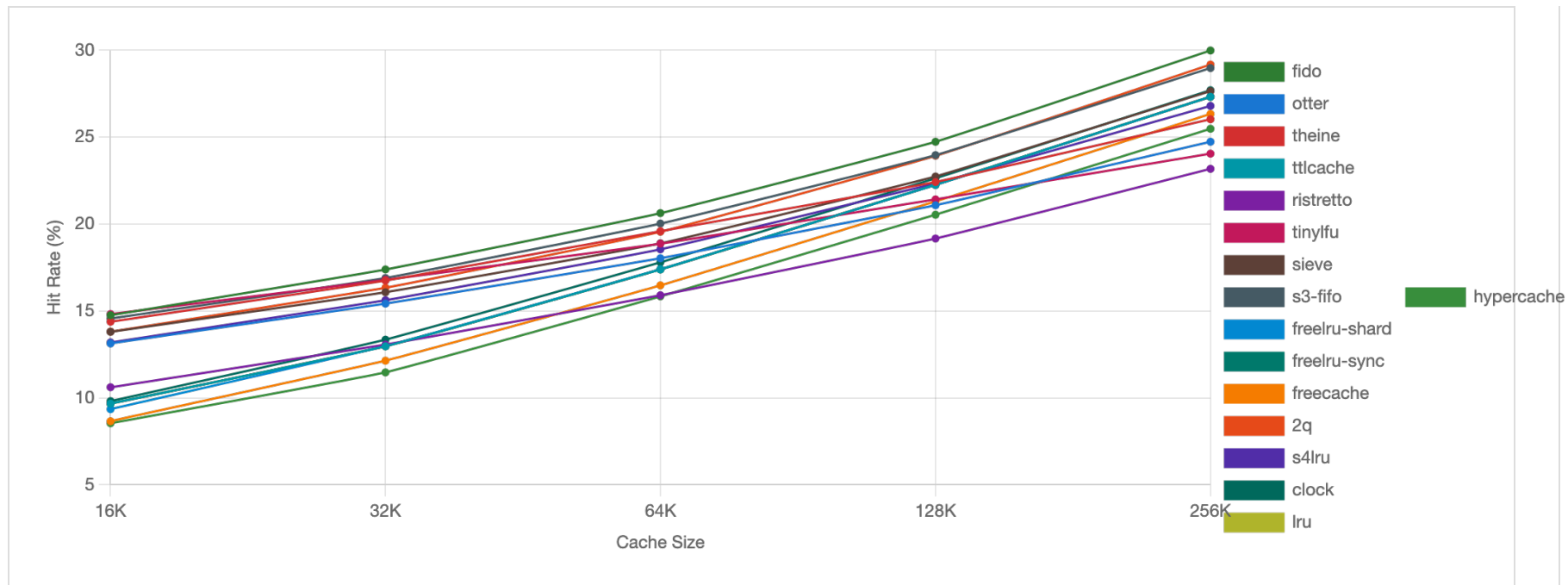
Results Table

| Cache     | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|-----------|---------|---------|---------|---------|---------|---------|
| freecache | 82.192% | 82.773% | 83.263% | 83.720% | 86.271% | 83.644% |
| 2q        | 82.451% | 83.197% | 83.840% | 84.082% | 84.344% | 83.583% |
| clock     | 82.576% | 83.041% | 83.446% | 83.826% | 84.344% | 83.446% |
| sieve     | 81.868% | 82.980% | 83.822% | 84.064% | 84.344% | 83.416% |
| s3-fifo   | 81.994% | 83.124% | 83.616% | 83.897% | 84.141% | 83.354% |
| fido      | 82.017% | 82.962% | 83.627% | 83.904% | 84.141% | 83.330% |
| s4lru     | 82.189% | 83.050% | 83.579% | 83.810% | 83.975% | 83.321% |
| tinylfu   | 80.956% | 81.574% | 83.225% | 84.927% | 85.916% | 83.320% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| lru           | 82.451% | 82.922% | 83.346% | 83.688% | 84.110% | 83.303% |
| ttlcache      | 82.451% | 82.922% | 83.346% | 83.688% | 84.110% | 83.303% |
| freelru-sync  | 82.451% | 82.922% | 83.346% | 83.688% | 84.110% | 83.303% |
| freelru-shard | 82.439% | 82.920% | 83.348% | 83.690% | 84.110% | 83.301% |
| hypercache    | 82.026% | 82.626% | 83.140% | 83.559% | 83.896% | 83.050% |
| theine        | 81.777% | 81.596% | 83.115% | 82.946% | 84.213% | 82.730% |
| otter         | 81.111% | 81.429% | 82.238% | 82.819% | 83.936% | 82.306% |
| ristretto     | 76.714% | 77.626% | 78.986% | 80.553% | 84.598% | 79.695% |

## Tencent Photo Trace

2M requests, ~1.34M unique photos. Source: [ATC '18](#)



Results Table

| Cache   | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------|---------|---------|---------|---------|---------|---------|
| fido    | 14.755% | 17.389% | 20.625% | 24.738% | 29.988% | 21.499% |
| s3-fifo | 14.570% | 16.903% | 20.031% | 23.965% | 28.982% | 20.890% |
| 2q      | 13.813% | 16.336% | 19.564% | 23.928% | 29.182% | 20.565% |
| theine  | 14.378% | 16.752% | 19.585% | 22.419% | 26.027% | 19.832% |
| sieve   | 13.795% | 16.084% | 18.895% | 22.736% | 27.648% | 19.831% |
| s4lru   | 13.187% | 15.624% | 18.551% | 22.347% | 26.802% | 19.302% |
| tinylfu | 14.817% | 16.786% | 18.872% | 21.420% | 24.051% | 19.189% |
| otter   | 13.127% | 15.423% | 18.034% | 21.078% | 24.727% | 18.478% |

| Cache         | 16K     | 32K     | 64K     | 128K    | 256K    | Avg     |
|---------------|---------|---------|---------|---------|---------|---------|
| clock         | 9.811%  | 13.352% | 17.799% | 22.631% | 27.696% | 18.258% |
| ttlcache      | 9.675%  | 12.972% | 17.390% | 22.264% | 27.319% | 17.924% |
| lru           | 9.675%  | 12.972% | 17.390% | 22.264% | 27.319% | 17.924% |
| freelru-sync  | 9.675%  | 12.972% | 17.390% | 22.264% | 27.319% | 17.924% |
| freelru-shard | 9.351%  | 12.978% | 17.399% | 22.254% | 27.318% | 17.860% |
| freecache     | 8.661%  | 12.139% | 16.468% | 21.303% | 26.327% | 16.979% |
| ristretto     | 10.610% | 13.072% | 15.911% | 19.170% | 23.176% | 16.388% |
| hypercache    | 8.539%  | 11.470% | 15.835% | 20.541% | 25.476% | 16.372% |

## Latency Benchmarks (Single-Threaded)



Lower latency means faster operations. Measures the time (in nanoseconds) for individual Get, Set, and SetEvict operations in a single-threaded environment.

### String Keys

#### Results

| Cache | Get (ns) | Get<br>allocs | Set (ns) | Set<br>allocs | SetEvict (ns) | SetEvict<br>allocs | Avg<br>(ns) |
|-------|----------|---------------|----------|---------------|---------------|--------------------|-------------|
|-------|----------|---------------|----------|---------------|---------------|--------------------|-------------|

| Cache             | Get (ns)                     | Get<br>allocs | Set (ns)                     | Set<br>allocs | SetEvict (ns)                | SetEvict<br>allocs | Avg<br>(ns) |
|-------------------|------------------------------|---------------|------------------------------|---------------|------------------------------|--------------------|-------------|
| fido              | <div><div></div></div> 10.0  | 0             | <div><div></div></div> 16.0  | 0             | <div><div></div></div> 119.0 | 1                  | 13.000      |
| clock             | <div><div></div></div> 16.0  | 0             | <div><div></div></div> 22.0  | 0             | <div><div></div></div> 98.0  | 2                  | 19.000      |
| s4lru             | <div><div></div></div> 18.0  | 0             | <div><div></div></div> 60.0  | 1             | <div><div></div></div> 62.0  | 1                  | 39.000      |
| sieve             | <div><div></div></div> 24.0  | 0             | <div><div></div></div> 46.0  | 0             | <div><div></div></div> 192.0 | 3                  | 35.000      |
| freelru-<br>sync  | <div><div></div></div> 24.0  | 0             | <div><div></div></div> 24.0  | 0             | <div><div></div></div> 36.0  | 0                  | 24.000      |
| freelru-<br>shard | <div><div></div></div> 25.0  | 0             | <div><div></div></div> 38.0  | 0             | <div><div></div></div> 47.0  | 0                  | 31.500      |
| 2q                | <div><div></div></div> 27.0  | 0             | <div><div></div></div> 30.0  | 0             | <div><div></div></div> 234.0 | 2                  | 28.500      |
| lru               | <div><div></div></div> 28.0  | 0             | <div><div></div></div> 30.0  | 0             | <div><div></div></div> 96.0  | 1                  | 29.000      |
| s3-fifo           | <div><div></div></div> 31.0  | 0             | <div><div></div></div> 46.0  | 0             | <div><div></div></div> 343.0 | 5                  | 38.500      |
| otter             | <div><div></div></div> 32.0  | 0             | <div><div></div></div> 134.0 | 1             | <div><div></div></div> 164.0 | 1                  | 83.000      |
| ristretto         | <div><div></div></div> 59.0  | 1             | <div><div></div></div> 262.0 | 3             | <div><div></div></div> 121.0 | 4                  | 160.500     |
| tinylfu           | <div><div></div></div> 83.0  | 0             | <div><div></div></div> 113.0 | 3             | <div><div></div></div> 119.0 | 3                  | 98.000      |
| freecache         | <div><div></div></div> 83.0  | 1             | <div><div></div></div> 58.0  | 0             | <div><div></div></div> 107.0 | 0                  | 70.500      |
| theine            | <div><div></div></div> 84.0  | 1             | <div><div></div></div> 114.0 | 0             | <div><div></div></div> 202.0 | 0                  | 99.000      |
| hypercache        | <div><div></div></div> 105.0 | 1             | <div><div></div></div> 215.0 | 2             | <div><div></div></div> 956.0 | 6                  | 160.000     |
| ttlcache          | <div><div></div></div> 461.0 | 0             | <div><div></div></div> 442.0 | 0             | <div><div></div></div> 368.0 | 3                  | 451.500     |

Int Keys

Results

| Cache ▾       | Get (ns) ▾                   | Get<br>allocs ▾ | Set (ns) ▾                   | Set<br>allocs ▾ | SetEvict (ns) ▾              | SetEvict<br>allocs ▾ | Avg<br>(ns) ▾ |
|---------------|------------------------------|-----------------|------------------------------|-----------------|------------------------------|----------------------|---------------|
| freelru-shard | <div><div></div></div> 7.0   | 0               | <div><div></div></div> 20.0  | 0               | <div><div></div></div> 18.0  | 0                    | 13.500        |
| fido          | <div><div></div></div> 8.0   | 0               | <div><div></div></div> 14.0  | 0               | <div><div></div></div> 100.0 | 1                    | 11.000        |
| otter         | <div><div></div></div> 32.0  | 0               | <div><div></div></div> 130.0 | 1               | <div><div></div></div> 155.0 | 1                    | 81.000        |
| theine        | <div><div></div></div> 79.0  | 1               | <div><div></div></div> 108.0 | 0               | <div><div></div></div> 198.0 | 0                    | 93.500        |
| ttlcache      | <div><div></div></div> 467.0 | 0               | <div><div></div></div> 433.0 | 0               | <div><div></div></div> 353.0 | 3                    | 450.000       |

String Keys - GetOrSet

GetOrSet is an atomic operation that gets a value if it exists, or sets it if it doesn't. Only caches that support this operation are shown.

GetOrSet Results

| Cache ▾    | GetOrSet (ns) ▾              | GetOrSet allocs ▾ |
|------------|------------------------------|-------------------|
| fido       | <div><div></div></div> 10.0  | 0                 |
| otter      | <div><div></div></div> 54.0  | 1                 |
| freecache  | <div><div></div></div> 125.0 | 2                 |
| hypercache | <div><div></div></div> 201.0 | 2                 |
| ttlcache   | <div><div></div></div> 487.0 | 0                 |

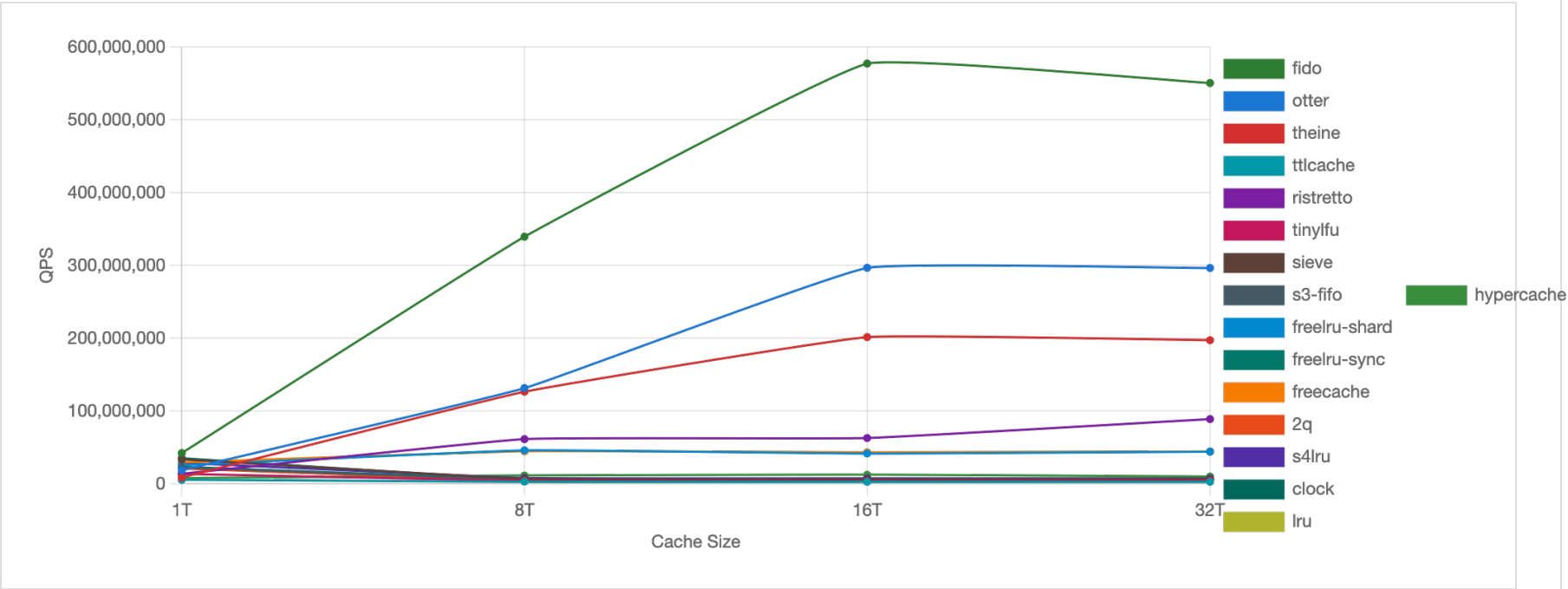


# Throughput Benchmarks (Multi-Threaded)



Higher throughput means more queries per second (QPS). Tests measure concurrent Get and Set performance separately with a Zipf workload across different thread counts.

## String Keys - Get

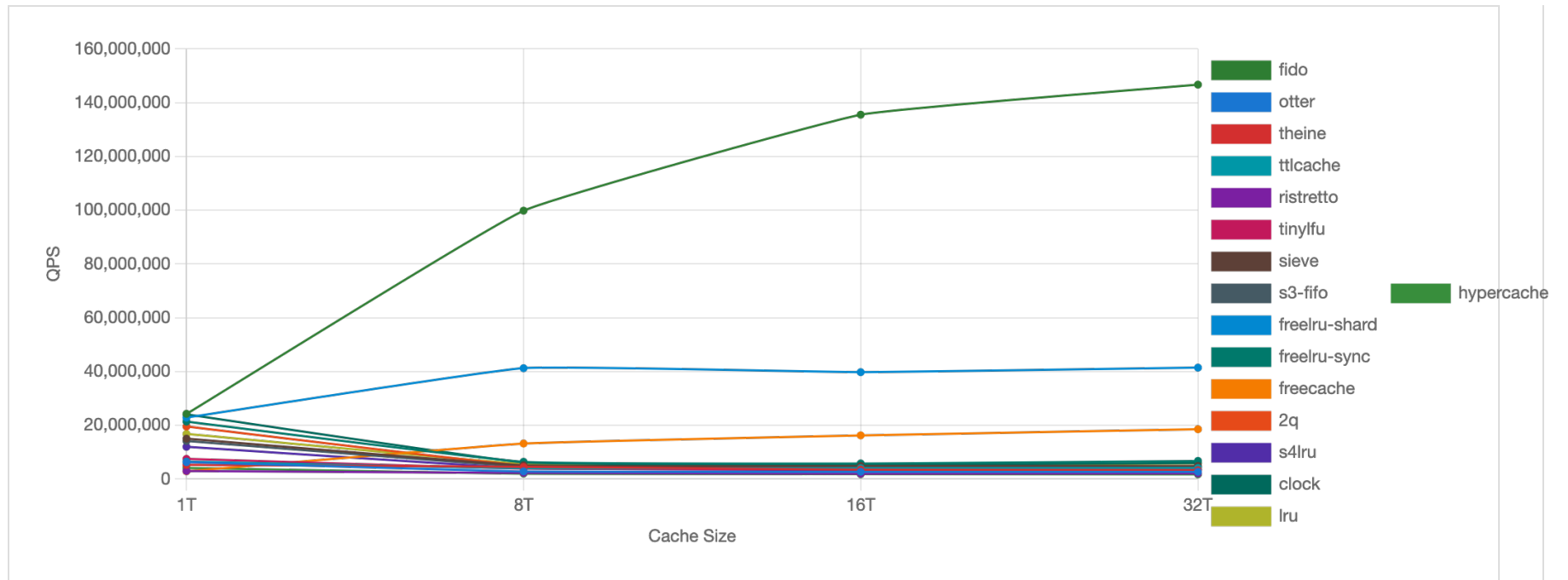


Results Table

| Cache | 1T     | 8T      | 16T     | 32T     | Avg     |
|-------|--------|---------|---------|---------|---------|
| fido  | 42.45M | 339.64M | 577.34M | 550.65M | 377.52M |

| Cache         | 1T     | 8T      | 16T     | 32T     | Avg     |
|---------------|--------|---------|---------|---------|---------|
| otter         | 18.89M | 131.65M | 296.77M | 296.50M | 185.95M |
| theine        | 8.80M  | 126.56M | 201.60M | 197.35M | 133.58M |
| ristretto     | 13.84M | 61.50M  | 62.97M  | 88.94M  | 56.81M  |
| freecache     | 29.59M | 44.90M  | 43.43M  | 44.54M  | 40.62M  |
| freelru-shard | 25.80M | 46.15M  | 41.75M  | 44.38M  | 39.52M  |
| sieve         | 33.63M | 7.10M   | 6.17M   | 6.78M   | 13.42M  |
| clock         | 35.20M | 6.11M   | 5.15M   | 6.13M   | 13.15M  |
| s4lru         | 28.91M | 8.23M   | 7.66M   | 7.67M   | 13.12M  |
| freelru-sync  | 23.43M | 6.90M   | 6.25M   | 7.48M   | 11.02M  |
| hypercache    | 7.90M  | 11.57M  | 12.60M  | 10.09M  | 10.54M  |
| s3-fifo       | 22.62M | 6.05M   | 5.41M   | 5.90M   | 9.99M   |
| lru           | 22.63M | 5.72M   | 5.06M   | 5.89M   | 9.82M   |
| 2q            | 21.02M | 4.75M   | 4.47M   | 5.24M   | 8.87M   |
| tinylfu       | 13.90M | 4.48M   | 3.92M   | 4.57M   | 6.72M   |
| ttlcache      | 5.80M  | 3.09M   | 2.95M   | 2.81M   | 3.66M   |

## String Keys - Set

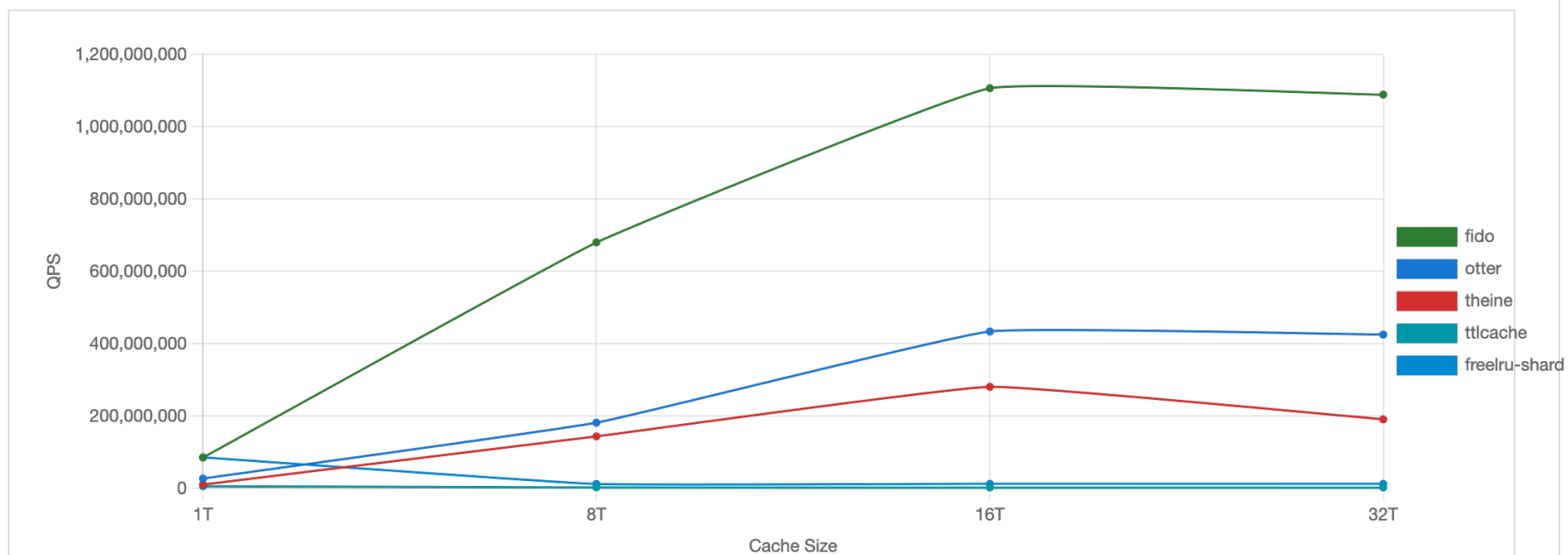


Results Table

| Cache         | 1T     | 8T     | 16T     | 32T     | Avg     |
|---------------|--------|--------|---------|---------|---------|
| fido          | 24.27M | 99.85M | 135.56M | 146.74M | 101.61M |
| freelru-shard | 22.89M | 41.28M | 39.78M  | 41.47M  | 36.35M  |
| freecache     | 2.92M  | 13.23M | 16.25M  | 18.57M  | 12.74M  |
| clock         | 24.06M | 6.24M  | 5.30M   | 5.96M   | 10.39M  |
| freelru-sync  | 21.41M | 6.43M  | 5.81M   | 6.75M   | 10.10M  |
| lru           | 16.84M | 5.69M  | 4.91M   | 5.89M   | 8.33M   |
| 2q            | 19.54M | 5.05M  | 4.23M   | 4.34M   | 8.29M   |
| sieve         | 15.07M | 5.15M  | 4.88M   | 4.92M   | 7.50M   |

| Cache      | 1T     | 8T    | 16T   | 32T   | Avg   |
|------------|--------|-------|-------|-------|-------|
| s3-fifo    | 14.21M | 4.75M | 4.32M | 4.63M | 6.97M |
| s4lru      | 12.05M | 4.10M | 3.40M | 3.60M | 5.79M |
| ttlcache   | 6.53M  | 4.06M | 4.11M | 4.15M | 4.71M |
| tinylfu    | 7.52M  | 3.90M | 3.64M | 3.50M | 4.64M |
| theine     | 5.30M  | 4.58M | 3.54M | 3.32M | 4.19M |
| otter      | 6.32M  | 2.85M | 2.70M | 2.70M | 3.64M |
| hypercache | 4.17M  | 2.04M | 1.87M | 1.79M | 2.47M |
| ristretto  | 3.01M  | 2.30M | 1.99M | 2.10M | 2.35M |

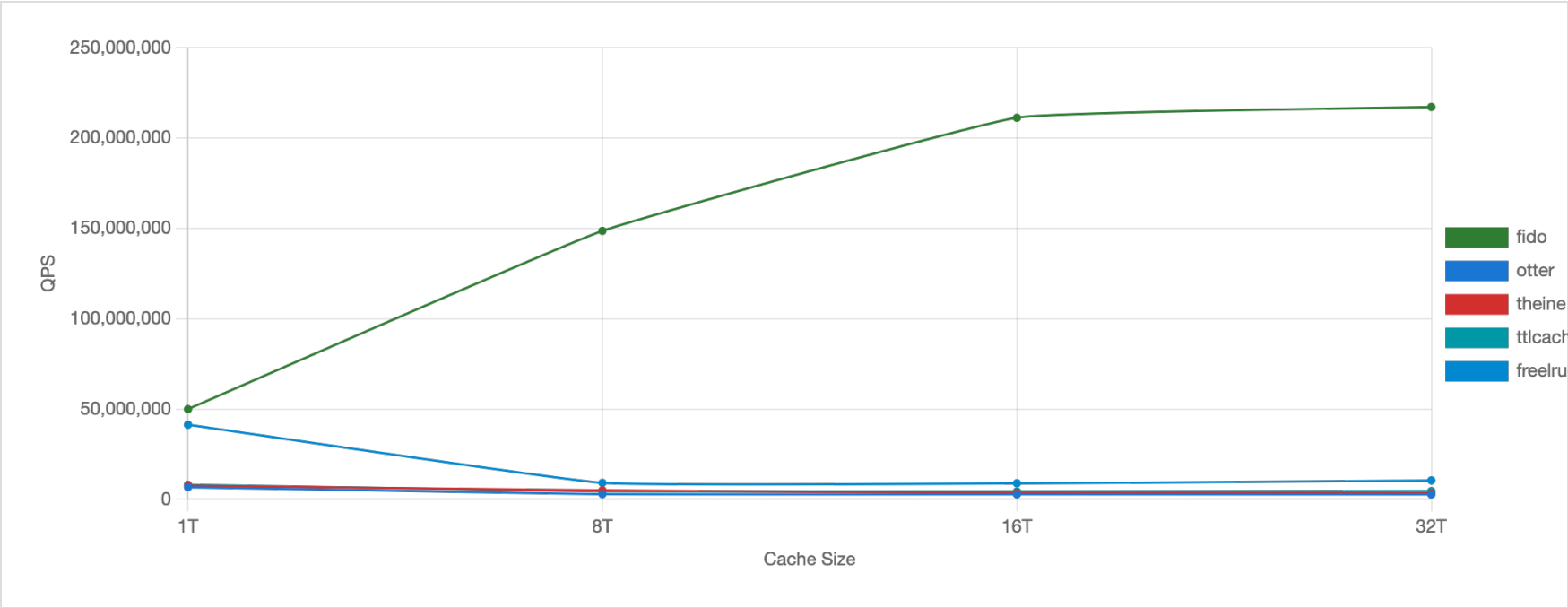
## Int Keys - Get



Results Table

| Cache         | 1T     | 8T      | 16T      | 32T      | Avg     |
|---------------|--------|---------|----------|----------|---------|
| fido          | 85.90M | 680.51M | 1107.06M | 1089.05M | 740.63M |
| otter         | 27.56M | 182.06M | 434.23M  | 425.75M  | 267.40M |
| theine        | 10.79M | 144.04M | 281.11M  | 191.25M  | 156.80M |
| freelru-shard | 86.33M | 12.65M  | 13.17M   | 13.08M   | 31.31M  |
| ttlcache      | 6.61M  | 3.20M   | 2.71M    | 2.61M    | 3.78M   |

Int Keys - Set

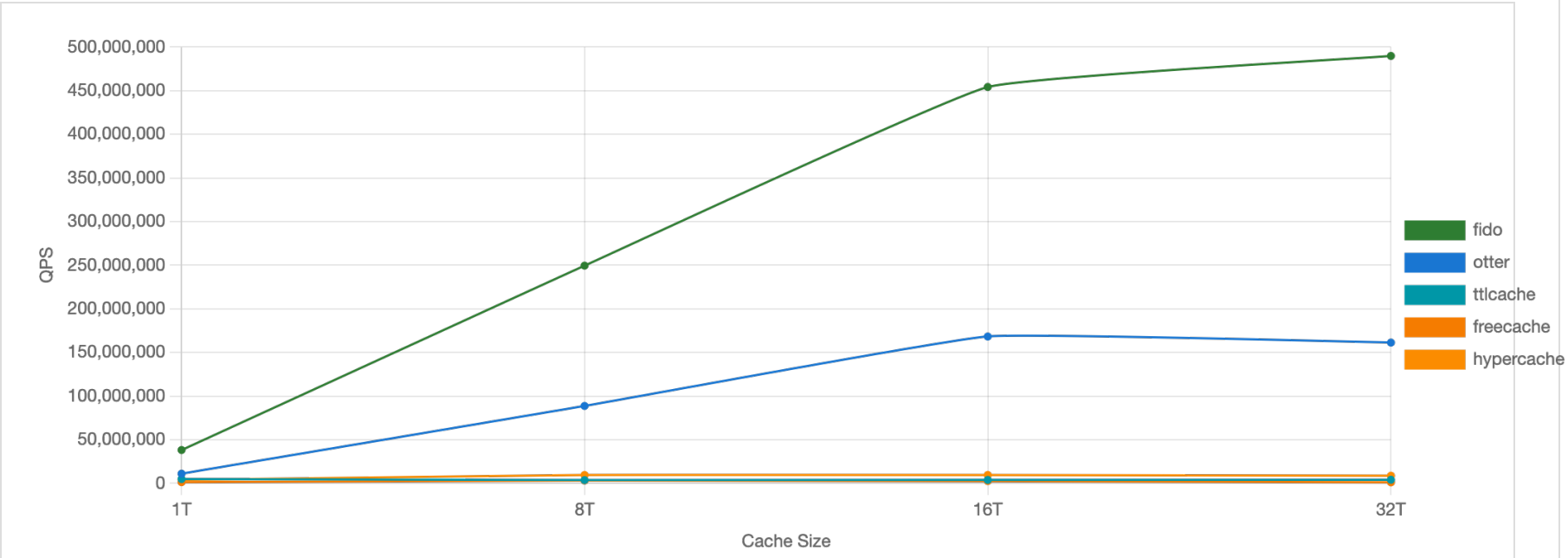


Results Table

| Cache         | 1T     | 8T      | 16T     | 32T     | Avg     |
|---------------|--------|---------|---------|---------|---------|
| fido          | 49.97M | 148.59M | 211.27M | 217.20M | 156.76M |
| freelru-shard | 41.33M | 9.12M   | 8.85M   | 10.43M  | 17.43M  |
| ttlcache      | 8.03M  | 4.43M   | 4.38M   | 4.60M   | 5.36M   |
| theine        | 7.56M  | 5.01M   | 3.49M   | 3.34M   | 4.85M   |
| otter         | 6.70M  | 2.90M   | 2.78M   | 2.74M   | 3.78M   |

GetOrSet Operations

Atomic get-or-compute operations with URL-like keys. Only caches supporting GetOrSet are shown.



Results Table

| Cache      | 1T     | 8T      | 16T     | 32T     | Avg     |
|------------|--------|---------|---------|---------|---------|
| fido       | 38.17M | 249.55M | 454.24M | 489.83M | 307.95M |
| otter      | 11.01M | 88.66M  | 168.34M | 161.40M | 107.35M |
| hypercache | 4.07M  | 9.58M   | 9.60M   | 8.82M   | 8.02M   |
| ttlcache   | 4.95M  | 3.77M   | 3.95M   | 4.06M   | 4.18M   |
| freecache  | 1.44M  | 3.23M   | 2.56M   | 1.28M   | 2.12M   |

Memory Overhead Benchmarks

Capacity: 32768 items, Value size: 1024 bytes. Measured in isolated processes. Overhead compared to baseline map[string][]byte. Lower is better.

Memory Results

| Cache        | Items Stored | Memory (MB) | Overhead vs map (bytes/item) |
|--------------|--------------|-------------|------------------------------|
| hypercache   | 0            | 0.31        | 0                            |
| freelru-sync | 32768        | 35.19       | -20                          |
| otter        | 32768        | 36.32       | 15                           |

| Cache         | Items Stored | Memory (MB) | Overhead vs map (bytes/item) |
|---------------|--------------|-------------|------------------------------|
| freelru-shard | 31606        | 36.47       | ■ 21                         |
| clock         | 32768        | 37.00       | ■ 37                         |
| fido          | 32768        | 37.38       | ■ 49                         |
| lru           | 32768        | 37.52       | ■ 54                         |
| 2q            | 32768        | 37.52       | ■ 54                         |
| s4lru         | 32768        | 38.27       | ■ 78                         |
| theine        | 32768        | 38.48       | ■ 84                         |
| s3-fifo       | 32768        | 39.54       | ■ 118                        |
| sieve         | 32768        | 39.54       | ■ 118                        |
| tinylfu       | 32767        | 39.83       | ■ 128                        |
| ttlcache      | 32768        | 40.82       | ■ 159                        |
| ristretto     | 32768        | 41.44       | ■ 179                        |
| freecache     | 32764        | 44.77       | ■ 286                        |